

Segment - 1

Integrity, Security & Relational Database Design:

↳ It is the practice of that data stored in the database, are accurate, consistent and free of errors.

Types of integrity constraints:

4 types.

① Domain constraints → column

Domain constraints not only allows us to test values inserted in the database, but also permits us to test queries to ensure that the comparisons made make sense.

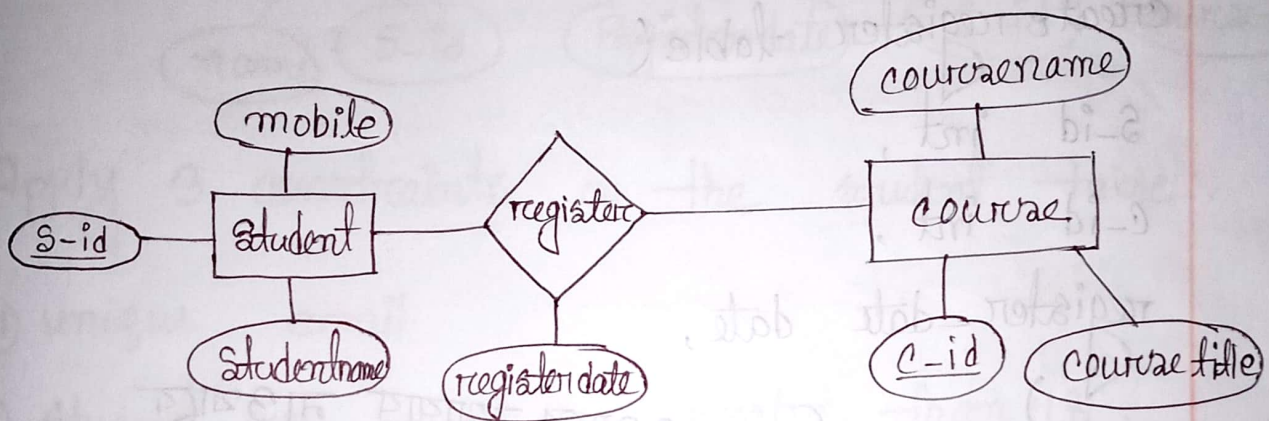
② Referential integrity constraint → table

referential integrity refers to the relationship between tables.

③ Assertion → overall database

An assertion is a predicate expressing a condition that we wish the database always to satisfy. Domain constraints and referential-integrity constraints are special forms of assertions.

iv) Triggers : A trigger is a statement that the system execute automatically as a side effect of a modification to the database.



create table student (
column name datatype
int, varchar, date, number

sid int primary key,
student name varchar(30)
not null

mobile number (11,0) unique
↓
decimal point এর আগে
বসে থাকবে
↓
decimal point এর
পরে
বসে থাকবে

```

create table course (
  cid int primary key
  course title varchar(20) not null
  course hrs numbers(2,0) not null
)

```

```

create register table (
  s_id int,
  c_id int,
  register_date date,
)

```

→ लिखा ७ था है खास नाउ था है

constraint fk foreign key (s_id) reference student (s_id)

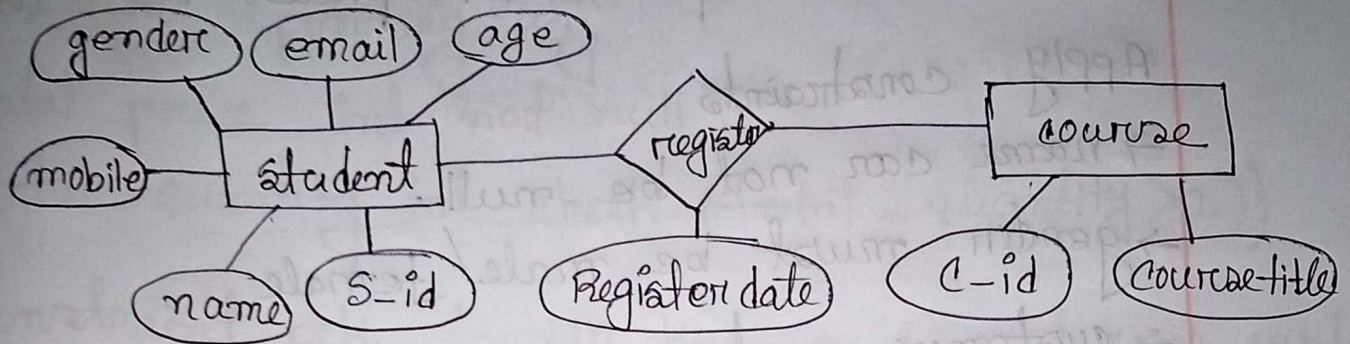
constraint fk foreign key (c_id) reference course (c_id)

```

create table student (
  s_id int,
  student name varchar(30) not null,
  ...
)

```


constraint p-sid primary key s-id.



Apply 3 constraints on the student table:

- i) unique email
- ii) student age must be greater than 18.
- iii) student gender must be male/female

⇒ create table student(
s-id int primary key,
name varchar(30) not null,
mobile number(11,0) unique,
email varchar(50) unique,
age int check (age > 18)
gender varchar(10) check (gender in ('male', 'female'))

customer (c_id, c_name, age, gender)

product (p_id, p_name, quantity)

purchase (c_id, p_id, total_price, total_quantity)

Apply constraints

→ name can not be null.

→ gender must be male/female

→ customer must purchase more than 1 product to purchase

⇒ create table customer (

c_id int primary key,

c_name varchar (30) not null,

age int,

gender varchar check (gender in ('male', 'female'))
)

create table product (

p_id int primary key,

p_name varchar (30) not null,

quantity int)


```

create table purchase (
  c-id int primary key,
  p-id int primary key,
  total price int not null,
  total quantity int check (total - quantity > 1))
constraint pck1 foreign key (c-id) reference customer (c-id)
constraint pck2 foreign key (p-id) reference product (p-id)

```

Assertion:

Condition we wish to database satisfy.

* Sum of all loan amount for each branch must be less than the sum of all account balance at the branch.

⇒ create assertion account constraint check
(not exist (select *
from branch

where (select sum (amount)
from loan

where loan . branch - name = branch . branch - name)

> = (select sum (amount)
from account

where account.branch_name = branch.branch_name)))

▣ Ensuring every loan customer keeps a minimum of \$1000 in an account.

⇒ Create assertion balance constraint check
not exists (select *
from loan
where not exists (select *
from borrower, depositor, account
where loan.loan_amount = borrow.loan_amount
and borrow.cname = depositor.cname and
depositor.account_num = account.account_num
and account.balance >= 1000))

▣ Trigger: It is a statement that is automatically executed by the system as a side effect of a modification to the database.

employee(e-id, name, address, position, e-id)

company(num-employee, revenue, e-id)

▣ Generate a trigger increment the number of employees each time new person is hired.

⇒ create trigger new-here
After insert an employee
for each row
update company
set num - employee = num -

set num = emp

Generate a trigger which decrement the number of employees each time an employee leaves the company.

⇒ create trigger vacancy
After delete an employee
for each row
update company
set num - employee = num

Suppose an overdraft is happened. Now create a trigger to overcome the overdraft by setting account balance zero and set the overdraft balance as loan.

⇒ overdraft ⇒ acc balance
100

withdraw
200

- 100

25

2154

balance

अपेक्ष overdraw

create trigger overdraft
After update on account
for each row referencing new row as new-row
new-row.balance < 0

BEGIN ATOMIC → keyword

insert into borrower values (select customer_name,
acc-number
from depositor
where depositor.acc-number
)

insert into loan values (new-rows, acc-number, new-
rows, branch-num, new-row, balance)

Update account

set balance = 0

where new-row.acc-num = account, acc-num

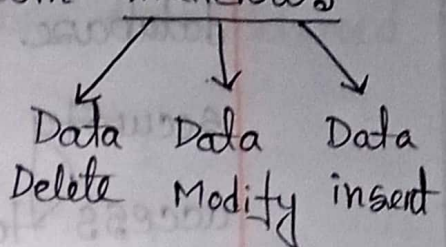
END

☐ Security:

Unauthorized user: System ko us access karne se
malicious access se unauthorized user ko us group se

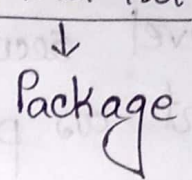
attack करे system को ग्रहण किए Intruders वृत्त

Security: It is a safe guarding from malicious access by intruders



```
graph TD; A[malicious] --> B[Data Delete]; A --> C[Data Modify]; A --> D[Data insert];
```

Malware: Virus, Worms, spyware, adware, ransomware, Trojan, rootkit, exploit kit



```
graph TD; A[exploit kit] --> B[Package];
```

Level of security measure:

- 1) Database
- 2) Operating system
- 3) Network
- 4) Physical
- 5) Human

1) Database: Some database-system users may be authorized to access only a limited portion of the database. Other users may be allowed to issue queries but may be forbidden to modify the data. It is responsibility of the database system to ensure that these

authorization restrictions are not violated.

2) **Operating system**: No matter how secure the database system is, weaknesses in operating-system security may serve as a means of unauthorized access to the database.

3) **Network**: Since almost all database systems allow remote access through terminals or networks, software-level security within the network software is as important as physical security, both on the Internet and in private networks.

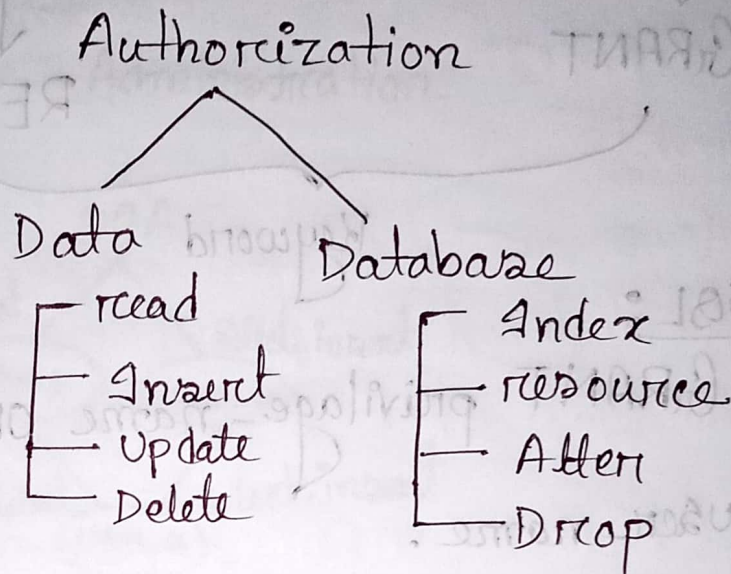
4) **Physical**: Sites with computer systems must be physically secured against armed or surreptitious entry by intruders.

5) **Human**: Users must be authorized carefully to reduce the chance of any user giving access to an intruder in exchange for a bribe or other favours.

Authorization :

In DBMS authorization refers to the process of granting or denying access and privileges

to users or roles. It is a crucial aspect of database security and helps control who can perform various operations on the data stored in the database.



Data:

Read: allows reading, but not modification of data.

Insert: allows insertion of new data but not modification of existing data.

Update: allows modification, but not deletion of data.

Delete: allows deletion of data.

Database:

Index: allows the creation and deletion of indices.

Resource: allows the creation of new relations.

Alter: allows the addition or deletion of attributes in a relation.

Drop: allows the deletion of relations.

DCL (Data control language)

GRANT

REVOKE

Keyword

SQL:

GRANT privilege-name ON table-name TO
User-name;

Ques: User 1 → employees → select, insert

⇒ GRANT select, insert ON employees TO user-1

Ques: User-1 user 2, then access rights

⇒ GRANT select, insert ON employees TO user 1 WITH
GRANT OPTION;

Keyword

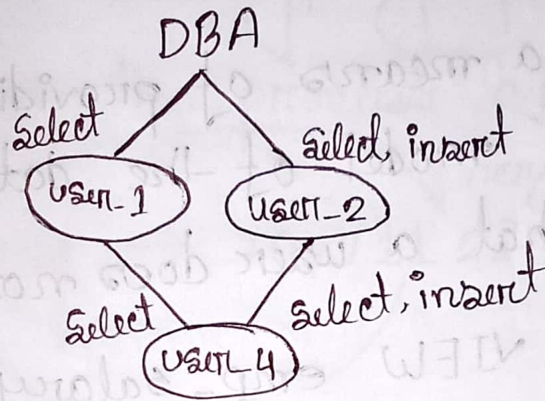
User - 2 then access (rights)

REVOKE privilege-name ON Table-name FROM user-

Ques: user 1 has insert privilege on employee table. How to revoke it?

⇒ REVOKE insert ON employees FROM user-1

DBA → Database Administration



Audit Trails: Many secure database applications require an audit trail be maintained. Audit trails is a chronological record of all the activities and changes made to a database.

User has security. So, selected privilege is not possible. Any privilege was not possible.

Role: User is not possible. Role is not possible.

CREATE ROLE amirrole

GRANT select ON employees TO amirrole

GRANT amirrole TO waer-1, waer-2

CREATE ROLE arcektarrole

GRANT amirrole TO arcektarrole ;

View:

Views as a means of providing a user with a personalized model of the database. A view can hide data that a user does not need to see.

CREATE VIEW emp-salary AS

(select *
from employees
where salary < 10000);

GRANT select ON emp-salary TO waer-1

GRANT select (name) ON emp-salary TO waer-1

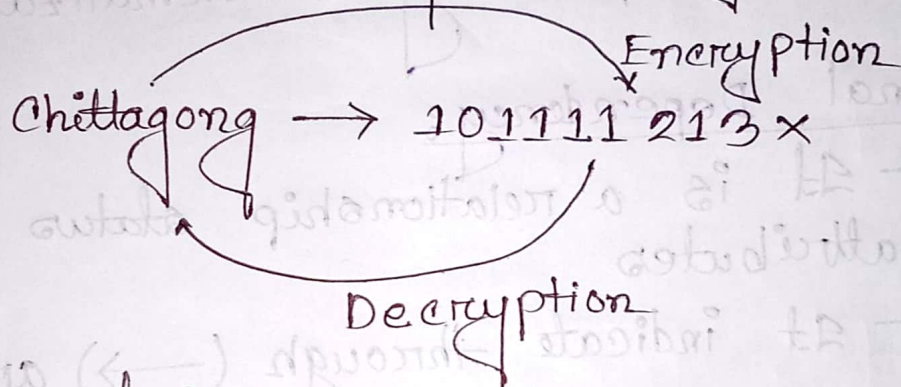
↓
column

↓
column

↓
एकल column (द्वारा जना)

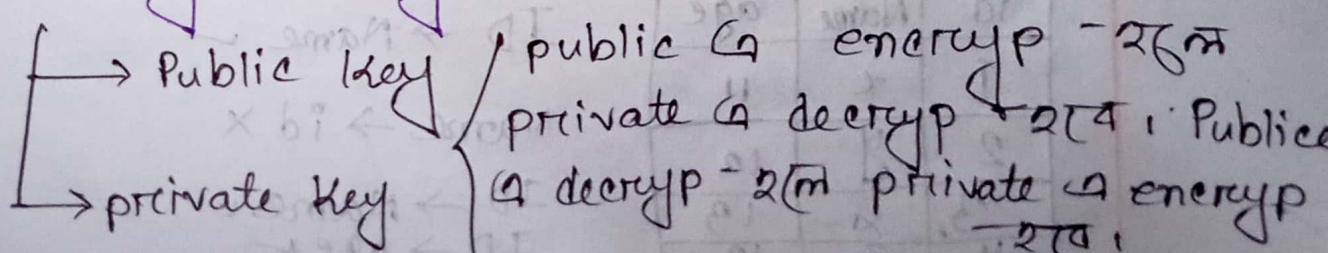
Encryption : Refers to the process of converting the data stored in a database into a secure, unreadable format using algorithms and encryption keys.

Decryption : A decryption key is a specific piece of information or cryptographic code that is used to reverse the process of encryption.



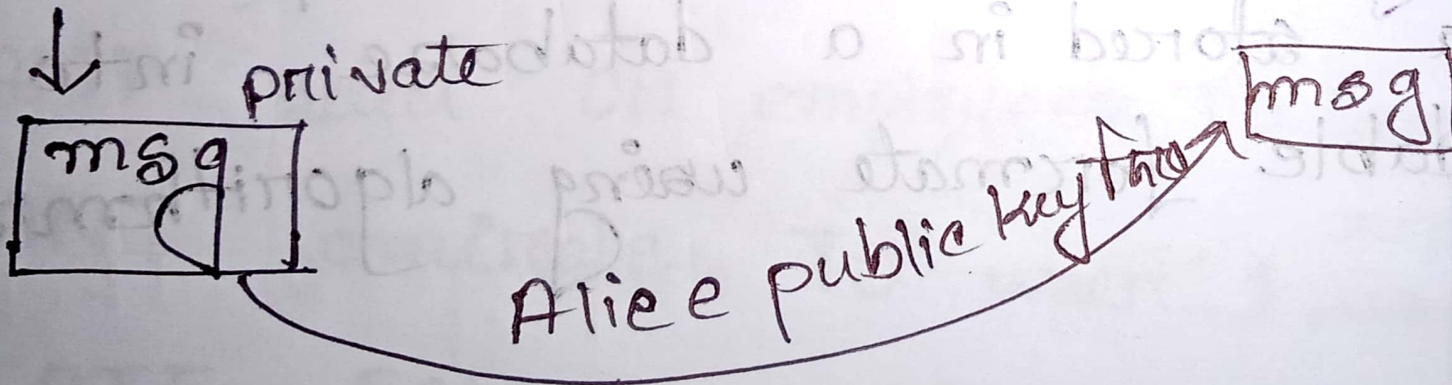
- Symmetric Key (Key same থাকবে)
- asymmetric Key (Key same থাকবে না)
- private Key
- public Key

Public key encryption :



Alice (private, public)

Bob (private, public)



Segment-2